Machine Learning

Assignment 3: Neural Network Generalization & Reliability

Submitted By

Lilian Ngozi Ugwu (NF1016558)

Submitted to: Professor - Ahmed Eltahawi

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University of Niagara Falls Canada

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Introduction

Health-risk prediction plays an important role in preventive healthcare, early intervention, and patient monitoring. The health_Risk_Tiers dataset provides a structured way to explore how neural networks behave when classifying individuals into four risk categories based on physiological and behavioral indicators. The purpose of this assignment is not simply to build a model that achieves high accuracy, but to understand how a neural network generalizes, how it fails, and how reliable its predictions are. To achieve this, two models were developed, a baseline Multilayer Perceptron (MLP) to observe natural overfitting behavior and a dropout-regularized MLP to examine how dropout improve generalization and stability.

Objectives

The objective of this assignment is to evaluate the model's generalization ability, identify overfitting, compare regularization strategies, analyze misclassification patterns, and assess the reliability of predicted probabilities.

Part A – Pre-Analysis & Expectations

Why Accuracy Is Misleading

Accuracy is not an appropriate primary metric for this dataset because the four classes are not equally represented. Tier 0 is the majority class, meaning a model could achieve deceptively high accuracy simply by predicting Tier 0 most of the time. In a healthcare context, this is dangerous misclassifying a high-risk patient (Tier 3) as low-risk can lead to delayed treatment and serious health consequences.

Metrics Prioritized

To ensure fair evaluation across all tiers, macro-averaged metrics which are Macro Precision, Macro Recall and Macro F1-Score were prioritized. **Justification:** These metrics treat each class equally and highlight performance on minority and high-risk tiers. They are more appropriate for health-risk prediction, where false negatives carry significant consequences.

Expected Overfitting

With 18 features and only 1,500 samples, the MLP is expected to be overfit. Typical signs are training loss decreases steadily, validation loss increasing and widening gap between training and validation

Overfitting Control

Dropout was selected as the regularization method. By randomly deactivating neurons during training, dropout reduces co-adaptation and encourages the model to learn more generalizable patterns.

Part B – Data Preparation

Missing Values

The dataset was checked for missing values, and none were found.

This allowed the analysis to proceed without imputation.

Class Imbalance: A count plot of the training labels confirmed that Tier 0 is the majority class. This imbalance reinforces the need for macro-averaged metrics and careful error analysis.

Train/Validation/Test Split

A stratified split was used to preserve class proportions, 70% training, 15% Validation and 15% test. This ensures that all tiers are represented consistently across splits.

Feature Scaling

StandardScaler was applied to all features. After scaling, the Mean of each feature was approximately zero and the Standard deviation was one. This step ensures stable gradient updates and prevents large-scale features such as triglycerides from dominating training.

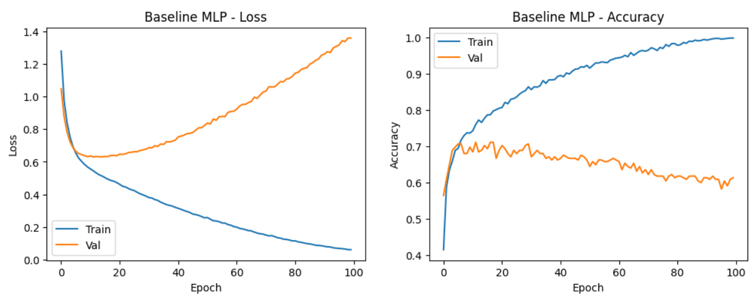
Part C – Baseline Neural Network

Model Architecture

The baseline MLP consisted of Dense (64, ReLU), Dense(32,ReLU) Dense(output_classes, Softmax) and no regularization was applied.

Training Behavior

Figure 1. Baseline MLP



To evaluate how well each model generalizes, I compared training vs validation loss and training vs validation accuracy across epochs.

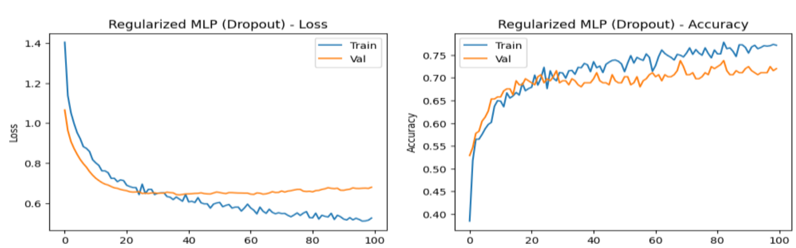
As shown in the learning curve figure 1, training loss decreased sharply, validation loss plateaued and then increased. Training accuracy became very high, and validation accuracy stagnated. There is large gap between training and validation loss which showed clear overfitting. This indicates that models memorized the training data rather than learning generalizable patterns.

Fit Assessment

Underfitting was Not observed - the model learned patterns well while Overfitting was Strongly present and Good Fit was Not achieved by the baseline model.

Part D – Controlled Regularization Experiment

Figure 2. Dropout – Regularized Model



A second MLP was trained with Dropout (0.4) added after each hidden layer. The architecture remained identical otherwise, ensuring a fair comparison.

Generalization Improvements

Compared to the baseline model as shown in fig.2, validation loss decreased early and stabilized. Validation accuracy remained stable around 0.70-0.72. The gap between training and validation accuracy was smaller. Although mild overfitting appeared in later epochs, the dropout-regularized model demonstrated significantly better generalization.

Overall Impact

Dropout successfully reduced overfitting and produced a more stable model. The regularized MLP achieved a better balance between bias and variance, making it more suitable for real-world prediction.

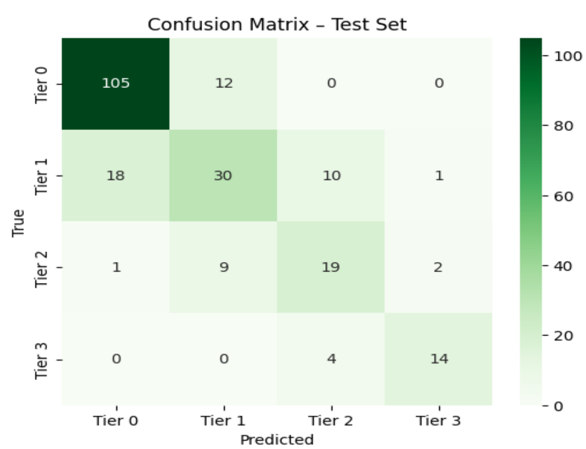
Part E – Final Evaluation & Error Analysis

Table 1. Final Model Performance Metrics

Metrics	Value
Macro Precision	0.709
Macro Recall	0.699
Macro F1-Score	0.703
Accuracy	0.747

As shown in table 1. The dropout regularized model achieved macro precision (0.709). macro-Recall (0.699). macro F1-Score (0.703) and accuracy of 0.747. These results confirm balanced performance across tiers

Figure 3. Confusion Matrix – Test Set



Error Analysis

The confusion matrix shows that Tier 1 and Tier 2 were the most difficult classes to classify, with Tier 1 often confused with Tier 0 or Tier 2 and Tier 2 frequently misclassified as Tier 1. Tier 3 was generally well identified but was occasionally predicted as Tier 2, which is clinically worrying. Overall, most errors occurred between adjacent tiers, and the model tended to under-predict the highest-risk class. A class weighted loss function would help address these issues by increasing the penalty for misclassifying minority and high-risk tiers, improving recall for Tier 2 and Tier 3

Part F – Model Reliability

Confidence vs Correctness

Table 2. Accuracy by Confidence Bin

Confidence Bin	Accuracy
Low confidence (0.0 – 0.5)	50%
Medium confidence (0.7- 0.85)	78%
High confidence (0.95 – 1.0)	100%

The results in table 2. show that higher confidence generally corresponds to higher correctness. As the model's confidence increases, its accuracy also improves. For example, prediction with low confidence have lower accuracy, while predictions with very high confidence like 0.95 – 1.0 are almost always correct. This suggests that the model is reasonably well calibrated.

Real-World Risk

one major of an overconfident model is that it may give very confident but incorrect predictions, which people may trust without questioning. For example, in a disease screening system, the model might predict with very high confidence that a patient is healthy when they have an early-stage illness. Because of this strong but wrong prediction, further tests might be skipped, leading to a missed diagnosis and delayed treatment, which can make the condition worse overtime.

Conclusion

The dropout regularized model demonstrated stronger generalization and more reliable performance than the baseline MLP, which suffered from significant overfitting. However, challenges in classifying Tier 1 and Tier 2 and the need for better calibration indicate that additional improvements such as class weighted loss and probability calibration techniques are required before the model can confidently applied in real clinical settings.

Recommendations

- Implement class-weighted loss to improve recall for high-risk tiers.
- Apply calibration techniques such as temperature scaling.
- Experiment with additional regularization (L2, early stopping).
- To reduce class imbalance, more Tier 1 and Tier 2 should be collected.

AI tools were used to clarify assignment instructions, assist with report structure and generate a basic coding template. All data preparation, model training, evaluation, error analysis, interpretation of results, and conclusions were completed by me without AI assistance.